

JACKSON CMELAK

Software Developer

Laguna Niguel, CA | (925) 858-4101 | jackowen@gmail.com | github.com/Monowog | linkedin.com/in/Jackson-Cmelak

U.S. Citizen | Eligible to Obtain Secret Clearance

EDUCATION

B.S. Computer Science
University of California, Davis

Sept 2022 - Dec 2025
Davis, CA

- Relevant coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Scientific Computation

EXPERIENCE

Freelance Web Developer
Self-Employed

Dec 2025 - Present
Laguna Niguel, CA

- Handle the full development lifecycle for production business websites built with Svelte and TypeScript, from requirements gathering through implementation, testing, deployment, and maintenance.
- Manage backend data storage and cloud hosting on Zoho Catalyst, including RESTful API and third-party SDK integrations.

Data Analyst Intern
US Cryotherapy

Jun 2022 - Aug 2022
San Ramon, CA

- Built regression models in Python (NumPy, Pandas) using five years of operational data to come up with recommendations for business stakeholders.
- Management adopted the recommendations, which shaped the following year's advertising strategy.

PROJECTS

UDP Telemetry Receiver
Personal Project

C++17 / CMake
Completed

- Wrote a C++17 service that parses a custom binary UDP packet format, checks checksums, and flags dropped or out-of-order packets from simulated GPS/IMU sensor data.
- Added a smoothing filter for noisy sensor readings, a synthetic packet sender for testing, and unit tests covering parsing, checksums, and sequence tracking.

WISH (Wisconsin Shell)
Operating Systems Course Project

C
Completed

- Wrote a Unix-style shell in C with built-in commands (cd, exit, path) that runs external programs using fork, exec, and wait.
- Added output redirection (>), parallel execution with & for running multiple commands at once, a batch mode for reading commands from a file, and custom error handling.

BoosterTutor
Real-Time AI Recommendation App

SvelteKit / TypeScript / PostgreSQL
In Development — Deployed on Vercel

- Building and deploying a full-stack SvelteKit app, from database schema through production deployment, that delivers real-time recommendations through the Claude API, implementing changes via the Linux (Mint) command line.
- Designed a PostgreSQL/Drizzle schema and ETL pipelines merging two datasets (Scryfall, 17lands) to ground LLM outputs in real data and cut down on hallucinated recommendations.

TECHNICAL SKILLS

Languages: C++, C, Python, TypeScript, SQL

Frameworks & Tools: Linux, Git, CMake, Svelte/SvelteKit, Node.js, PostgreSQL, SQLite3, Drizzle ORM, NumPy, Pandas, OpenAPI, Vercel

Practices: Systems Programming, RESTful API Design, Unit Testing, ETL / Data Pipelines, SDK Integration, LLM API Integration, Agentic Coding Workflows